```
    Flatten(),
    Dense(128, activation='relu'),
    Dropout(0.5),
    Dense(10, activation='softmax')
])

# Step 4: Compile Model
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy', metrics=['accuracy'])

# Step 5: Train Model
model.fit(datagen.flow(train_images, train_labels, batch_size=64), epochs=10, validation_data=
(test_images, test_labels)) # Step 6: Evaluate Model
test_loss, test_accuracy = model.evaluate(test_images, test_labels) print(f"Test Accuracy:
{test_accuracy * 100:.2f}%") # Step 7: Save and Deploy Model
model.save('image_classification_model.h5')
```

Project 5: Text Summarization with NLP

Objective:

To create a tool that generates concise and coherent summaries from lengthy documents, articles, or other textual content. The project aims to save users time by distilling key information and presenting it in a clear and understandable format. Additionally, the project emphasizes providing users with customizable summarization options, enhancing usability across diverse domains.

Detailed Plan:

1. Text Preprocessing:
    - Use natural language processing (NLP) libraries like NLTK or spaCy to preprocess the text.
    - Tasks include:
        - Tokenizing sentences and words.
        - Removing stopwords, punctuation, and irrelevant characters.
        - Converting all text to lowercase for consistency.
        - Lemmatization or stemming to standardize word forms.